

Explainable AI Methods

Matthew Engelhard

Today

- Contrastive learning on the board (SimCLR and CLIP batches and losses)
1. Interpretable versus explainable (definition, example)
 2. Competing perspectives on interpretability
 3. Evolving views of interpretability
 4. Simple explanation methods (saliency maps, Grad-CAM)
 5. More principled explanation methods (integrated & expected gradients, SHAP)

Interpretable vs Explainable

Interpretable versus explainable models

An interpretable model:

It is easy to for us to understand why the model makes the predictions that it makes.

An explainable model:

One or more techniques can be used to provide a human-friendly explanation for each model prediction.

Interpretable

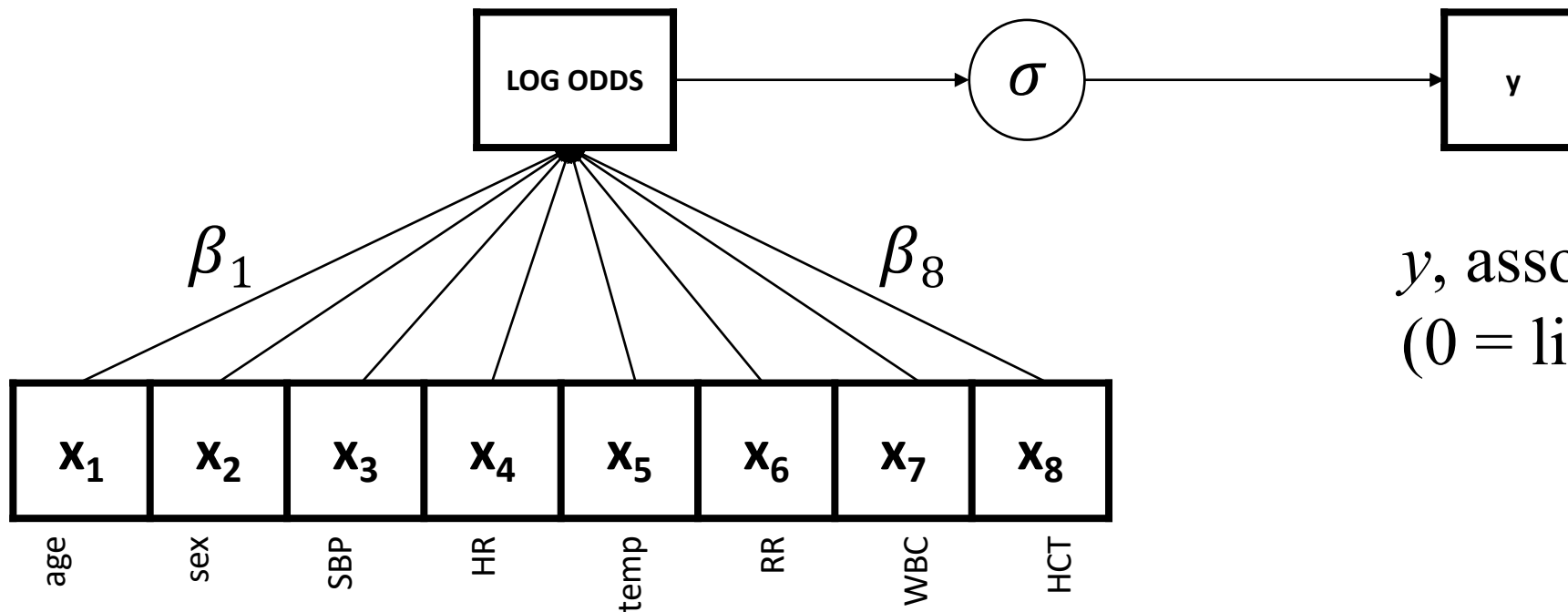


Explainable



Interpretable: logistic regression

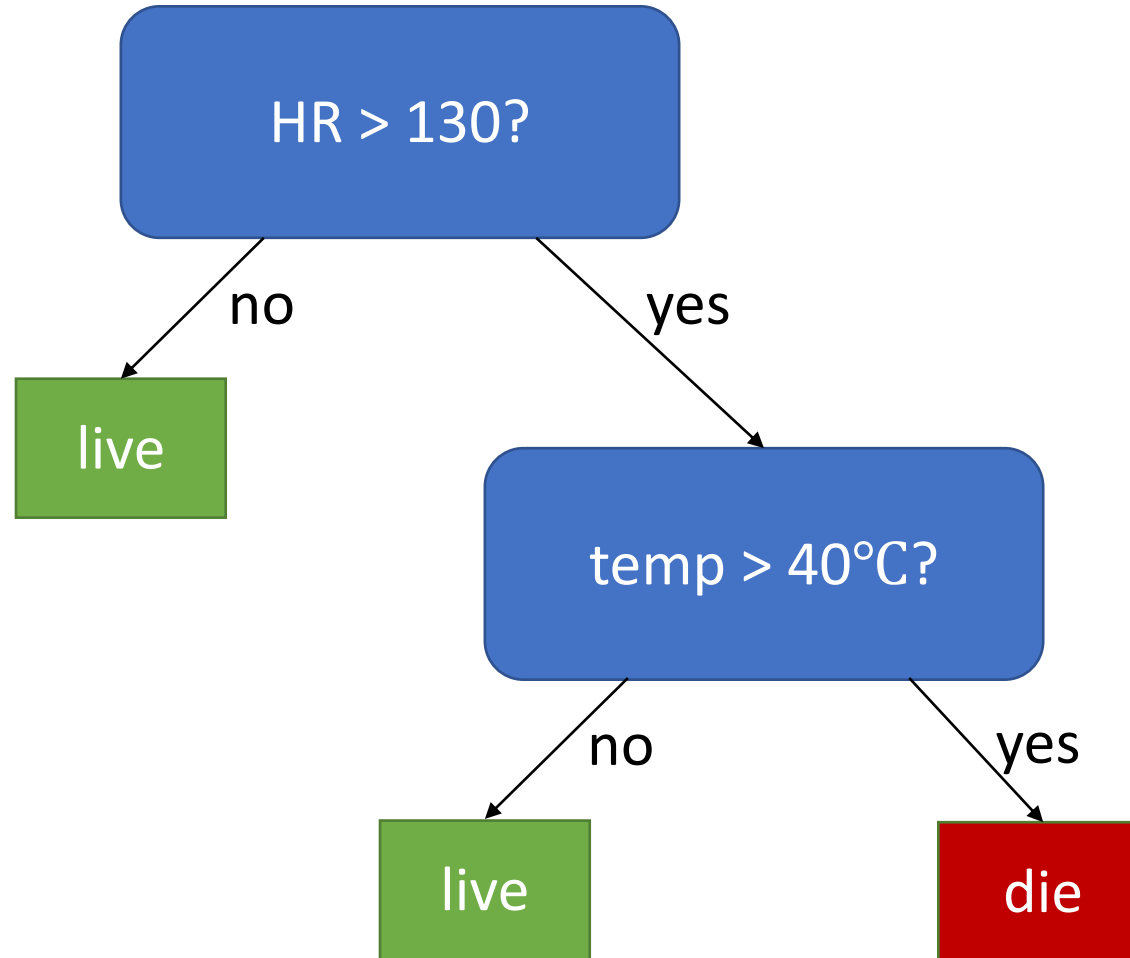
- Suppose a new patient is transferred to the ICU, and the (logistic regression) model predicts their mortality risk is high.
- **Q:** Is it hard to figure out why it made that prediction?
- **A:** No. You can look at the coefficients to see which variables increased and decreased the predicted probability.



y , associated label:
(0 = live, 1 = die)

Other Interpretable Models?

- Decision Tree



Explainable: CNN

- Suppose you apply a CNN to images of animals, and the model predicts that an animal is a meerkat.
- **Q:** Is it hard to figure out why it made that prediction?
- **A:** Yes, an explanation can be provided:



- Similarly, we can identify passages within a text document that would change predictions most if they were removed.

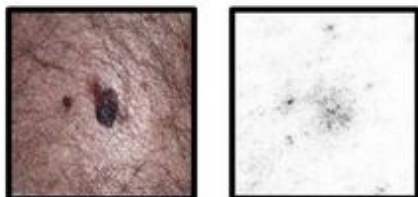
Caveats:

- This is for a single example; not clear how to summarize across all examples for images and text
- Can be very computationally expensive

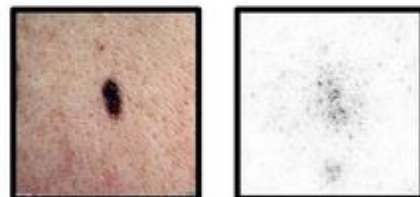


Explainable: CNN

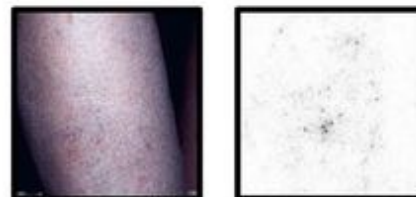
a. Malignant Melanocytic Lesion



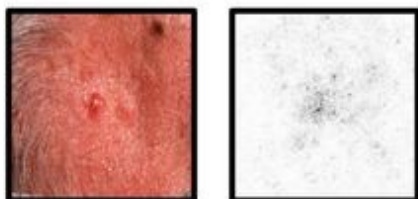
d. Benign Melanocytic Lesion



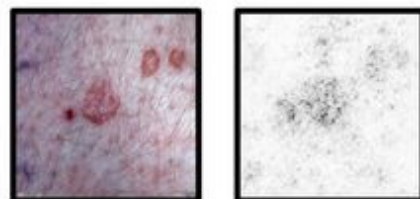
g. Inflammatory Condition



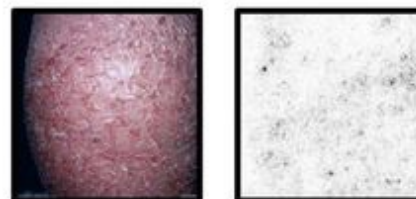
b. Malignant Epidermal Lesion



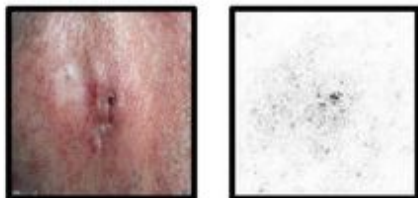
e. Benign Epidermal Lesion



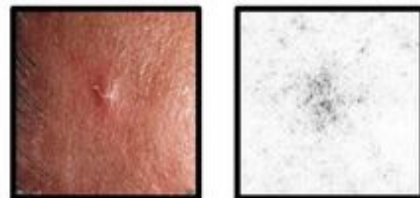
h. Genodermatosis



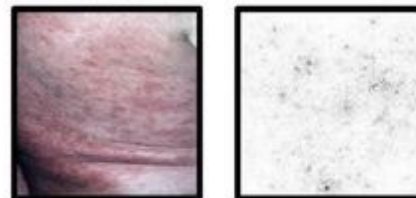
c. Malignant Dermal Lesion



f. Benign Dermal Lesion



i. Cutaneous Lymphoma



Saliency maps show gradients for each pixel with respect to the CNN's loss function. Darker pixels represent those with more influence.

How would this compare to a dermatologist's explanation?

Interpretable versus explainable models

An interpretable model:

We can understand how the model works *in general*, for example by inspecting coefficients or tree structure.

An explainable model:

We are presented with *case-by-case explanations* that decompose predictions into the contributions of different input features.

Competing perspectives on interpretability

Interpretable

?

Explainable

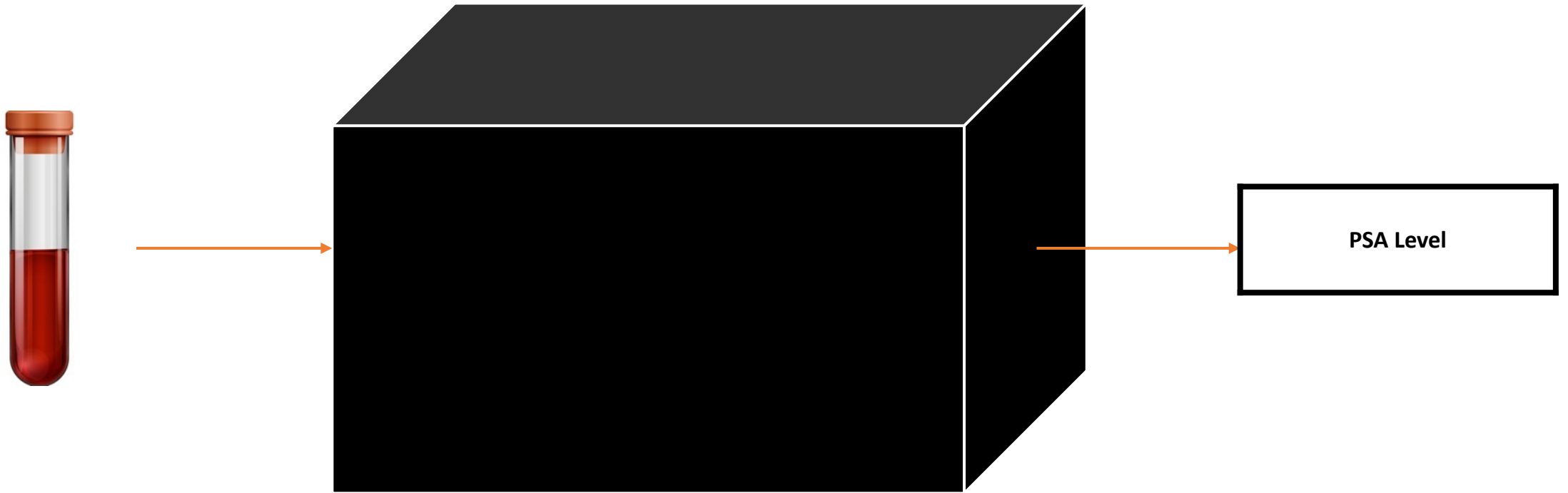
where does clinician decision-making fall on this spectrum?



Compared to simpler models, DNNs are a black box.
We can explain model predictions, but it's case by case.



Is PSA measurement a black box?



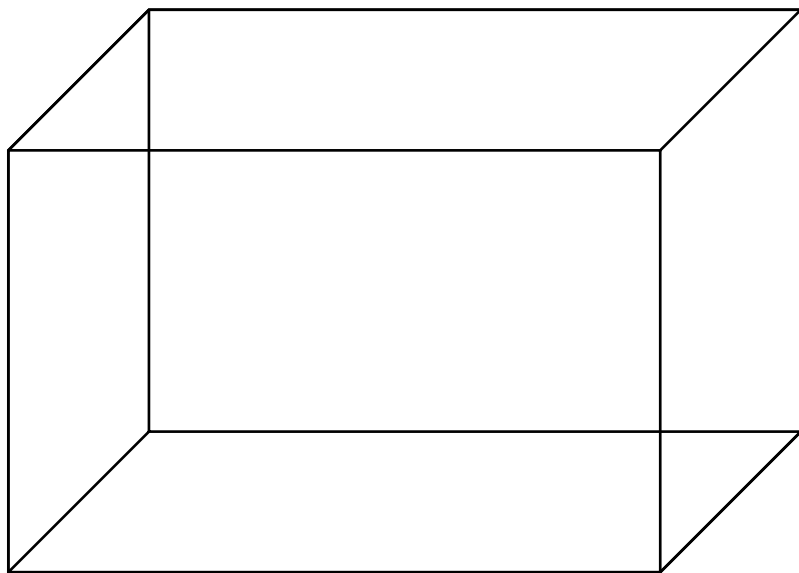
Is clinician decision-making a black box?



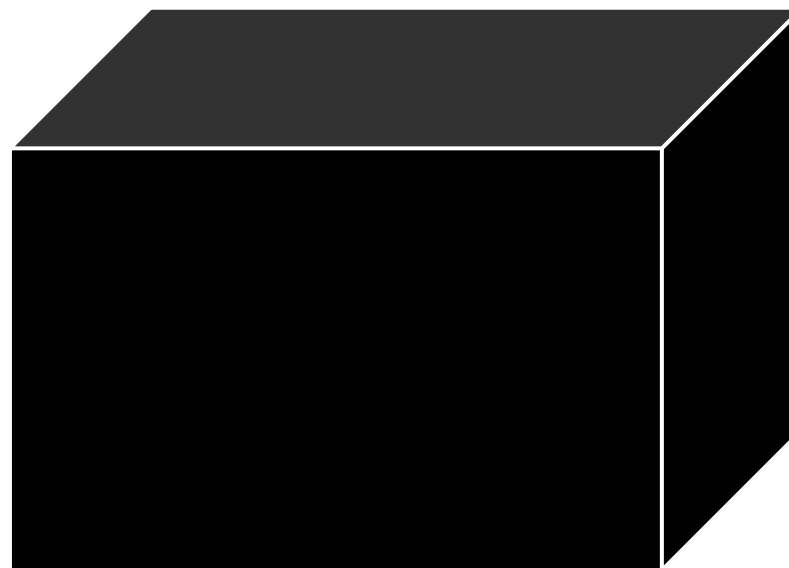
Two competing perspectives

FDA regulation of clinical decision support considers interpretability.

We can only use tools if we fully understand how they work



We just need to make sure our tools are *valid* and *reliable*



Evolving views of interpretability

Evolving views of interpretability

- **Old definition:** It is easy to for us to understand why the model makes the predictions that it makes.
- **New definition:** Predictions are explained based on elements or concepts that are understandable to humans.

This looks like that

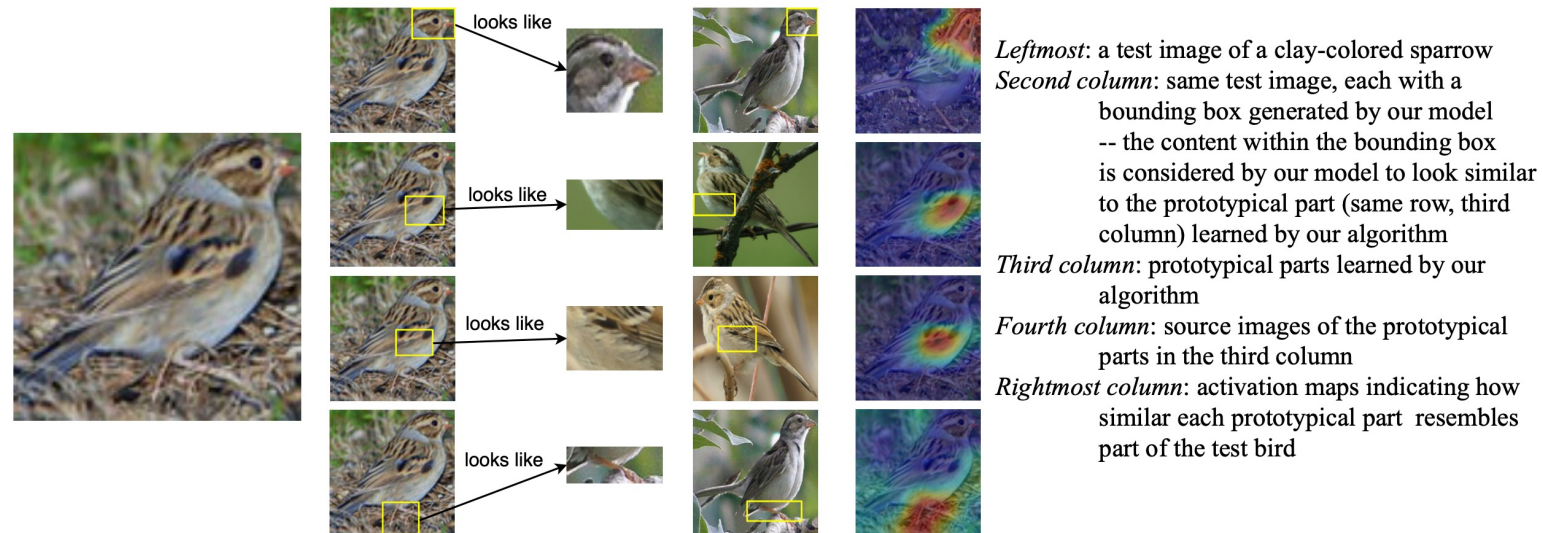


Figure 1: Image of a clay colored sparrow and how parts of it look like some learned prototypical parts of a clay colored sparrow used to classify the bird's species.

Uses large convolutional filters that can then be used to retrieve best-matching patches from the training set.

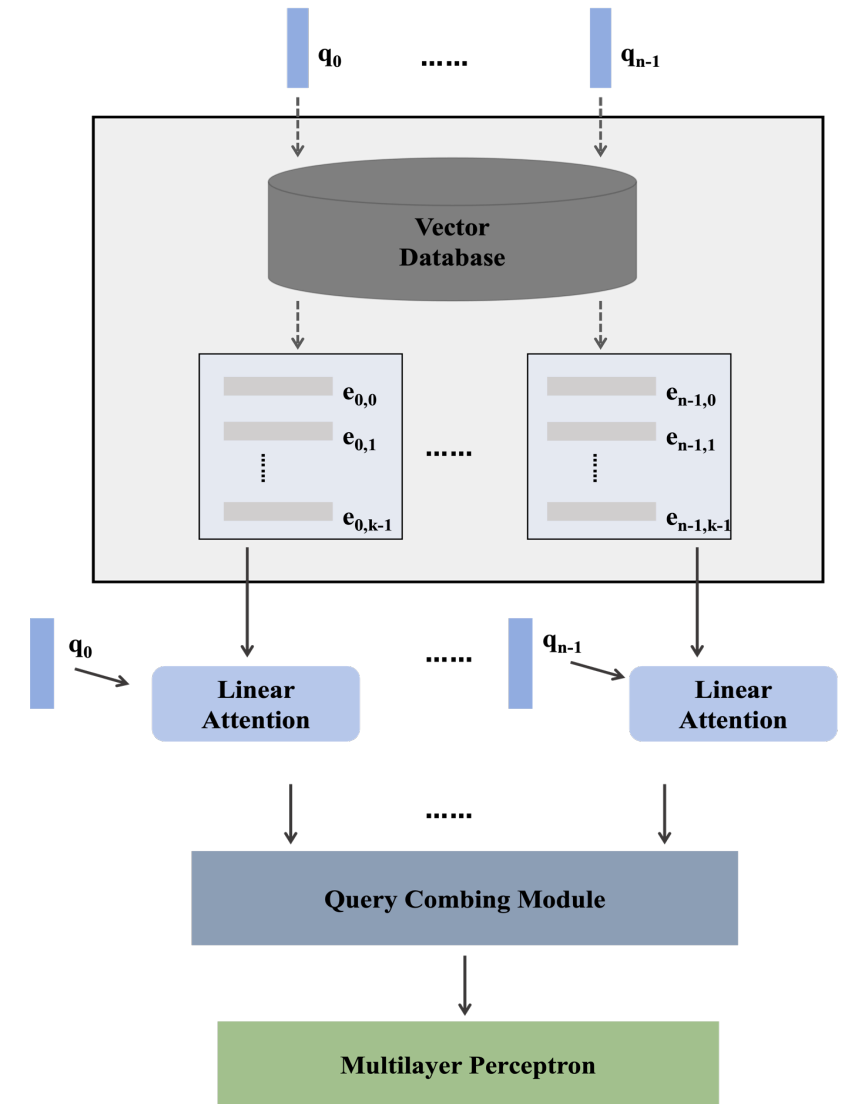
Justify prediction from evidence

Our approach (IRIS):

- Rather than manually defining queries using prior knowledge, the model learns query vectors directly from data.

Model Structure

- The model has n query vectors, initialized randomly and optimized as learnable parameters
- Each query vector retrieves k relevant clinical note embeddings. Only these retrieved embeddings are used for downstream risk prediction
- A key mechanism aggregates the retrieved embeddings, enabling the dynamic refinement of the query vectors during training.
- Specially designed tricks encourage each query to focus on a distinct, interpretable classification factor



Fengnan Li, Elliot Hill, Jiang Shu, Jiaxin Gao, and Matthew Engelhard. IRIS: Interpretable Retrieval-Augmented Classification for Long Interspersed Document Sequences. In: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025).

Justify prediction from evidence

Query Interpretation	Typical content of retrieved notes
Skill development	Assessment of developmental milestones including motor skills
Ophthalmological findings*	Findings such as esotropia, exotropia, amblyopia, myopia
Referral to specialty care	Referral to services including ophthalmology and neurology
Late talking	Speech history provided by parent, including current vocabulary
Birth measurements	Percentiles for birth weight, length, and head circumference

To identify which risk factors the model recognizes, We extract query vectors from the trained model and use them to retrieve the most relevant clinical note segments from the vector database. Categories of descriptive findings predictive of autism, as identified by IRIS, are listed in the table.

*consistent with e.g. Perna et al., Association between Autism Spectrum Disorder (ASD) and vision problems. A systematic review and meta-analysis. Mol Psychiatry. 2023 Dec;28(12):5011-5023.

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Simple explanation methods

Saliency maps

For model $f: \mathcal{X} \rightarrow (0,1)$, the saliency map $S(x)$ is given by:

$$S(x) = |\nabla_x f(x)|.$$

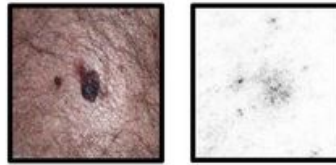
The importance of pixel i is:

$$S_i(x) = \left| \frac{\partial f(x)}{\partial x_i} \right|$$

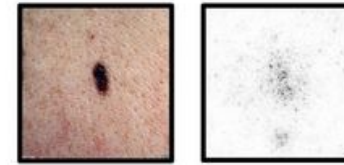
Intuition:

“How fast does the prediction change as I change pixel i ?”

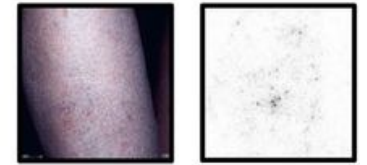
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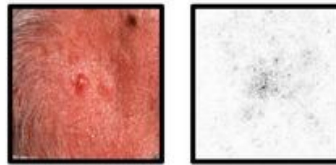
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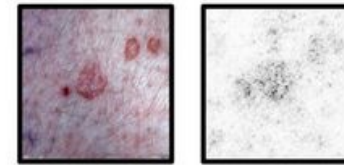
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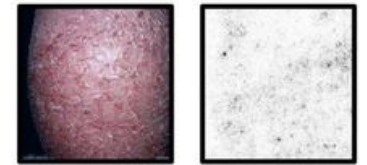
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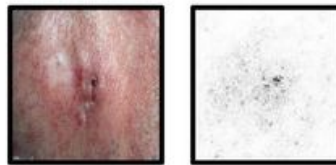
e. Benign Epidermal Lesion



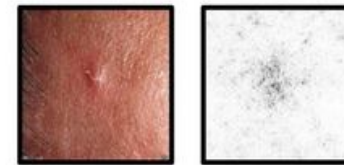
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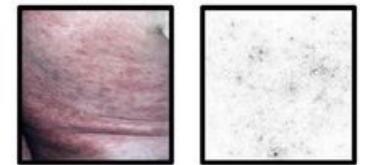
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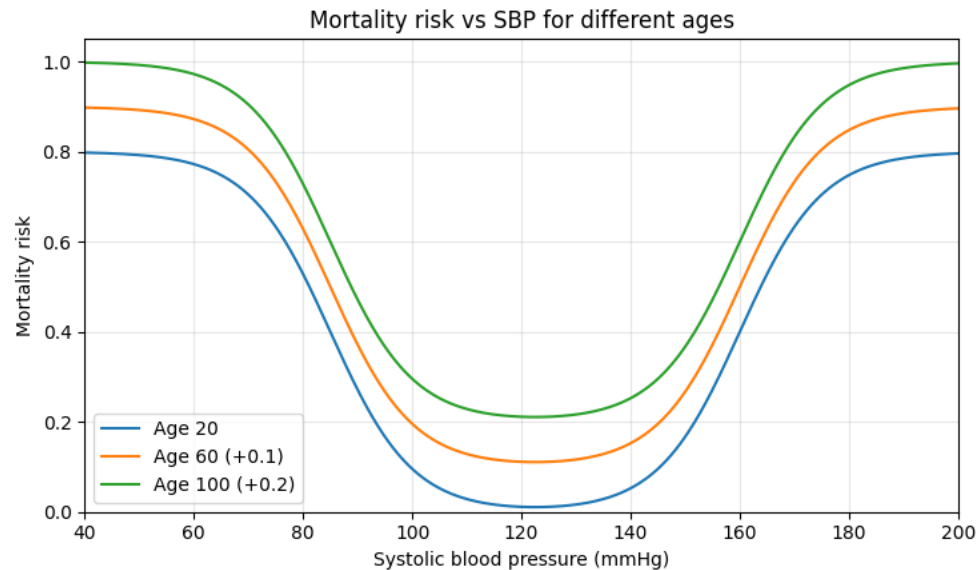


i. Cutaneous Lymphoma



Esteva et al., 2017

What's wrong with saliency maps?



Suppose we have the following model predicting mortality risk:

$$f(\text{SBP}, \text{AGE}) = \frac{\text{AGE}-20}{400} + \sigma\left(\frac{85-\text{SBP}}{7.5}\right) + \sigma\left(\frac{\text{SBP}-160}{7.5}\right)$$

$$S_{\text{SBP}}(\text{SBP}, \text{AGE}) = \left| \frac{\partial f(\text{SBP}, \text{AGE})}{\partial \text{SBP}} \right|$$

Consider the saliency map for a 60 y.o. with SBP=200.

- What is s_{SBP} ?
- What is s_{AGE} ?

Is this an accurate assessment of importance?

Grad-CAM

Similar to saliency maps, but instead of the raw input we use the last convolutional feature maps, which gives a smoother localization map.

1. Compute importance weights

$$\alpha_k = \text{mean}_{i,j} \left(\frac{\partial f(A_k(x))}{\partial A_{i,j}^k(x)} \right)$$

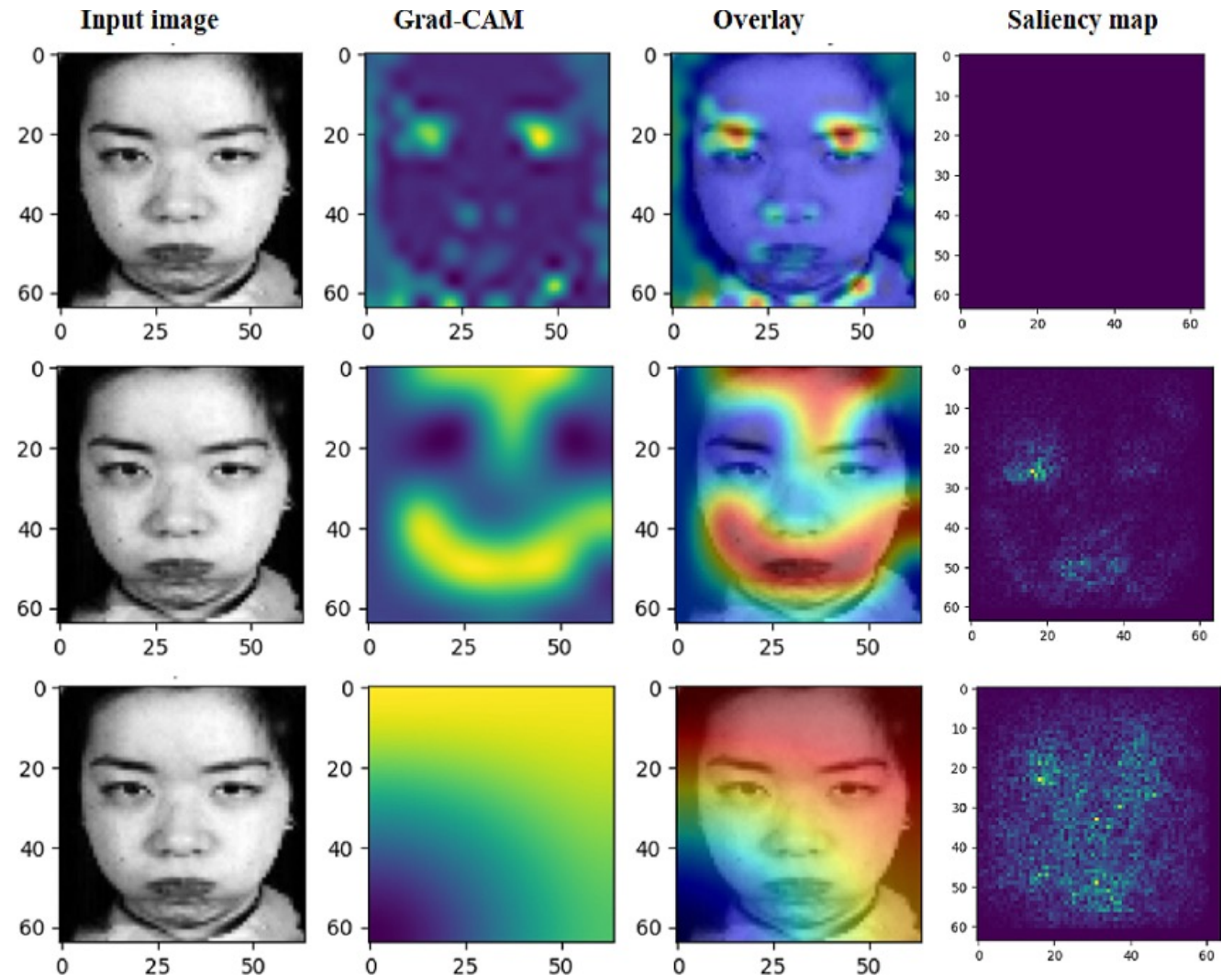
where $A_k(x)$ is the k th feature map and i, j are spatial coordinates

2. Grad-CAM is then a weighted combination of the feature maps

$$\phi^{GC}(x) = \text{ReLU} \left(\sum_k \alpha_k A^k(x) \right)$$

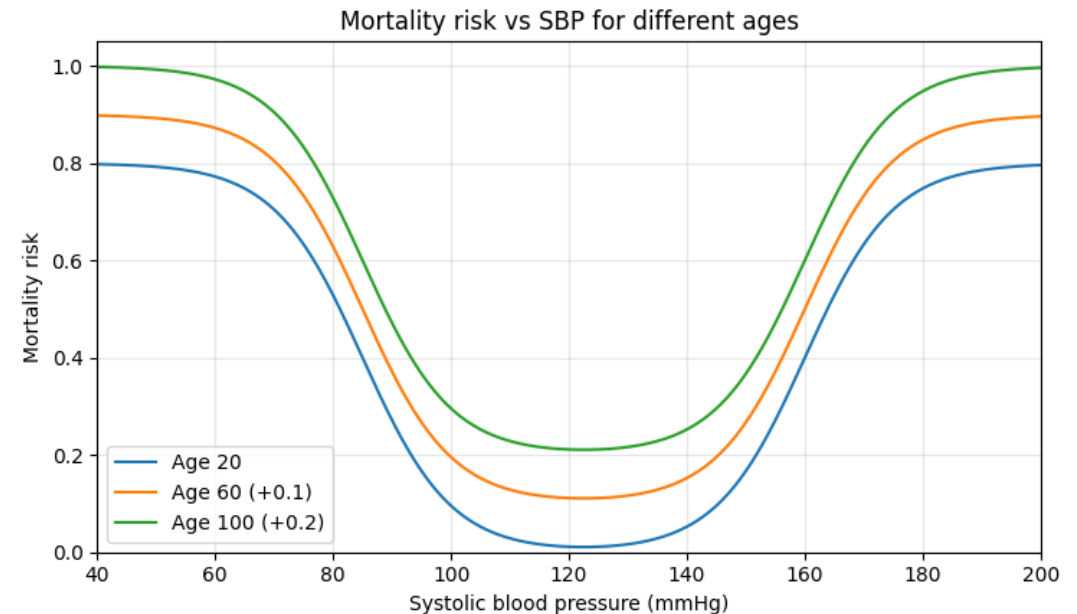
Grad-CAM versus saliency maps for emotion recognition

Grad-CAM and Saliency maps for different CNN architectures (16, 32), (32,64, 128), (16, 32, 64, 128) layers for angry emotion



What's wrong with Grad-CAM?

- The limitation is the same as saliency maps
- In short, this still focuses on the gradient evaluated at the current input which doesn't always accurately assess the overall importance of a given feature or region



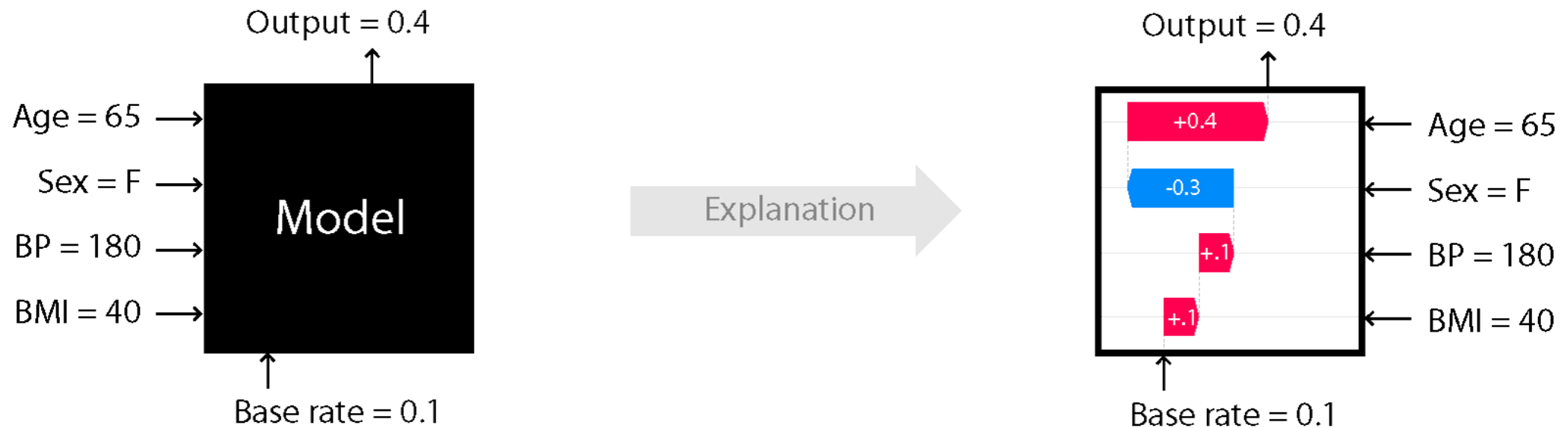
Better explanation methods

Desirable properties of feature attributions

Consider feature attributions $\phi_i(f, x, x')$ for model f , input x , and a reference input x' .

- Sensitivity: if x and x' differ in exactly one feature i , and $f(x) \neq f(x')$, then the attribution $\phi_i(f, x, x')$ should be nonzero.
- Implementation invariance: if $f=g$ pointwise, then $\phi(f, x, x') = \phi(g, x, x')$ for all x .
- Completeness (i.e. conservation): The sum of attributions across all features should satisfy $\sum_i \phi_i(f, x, x') = f(x) - f(x')$.

Desirable properties of feature attributions



Goal: decompose $f(x) - f(x')$ into the additive contributions of individual features.

A simple proposal

Define x^{-i} by setting the i th feature of x to the i th feature of x' and consider $\phi_i(f, x, x') = f(x) - f(x^{-i})$. Note that $x^{-i} = x + (x' - x)e_i$.

Does this satisfy the completeness axiom? $\sum_i \phi_i(f, x, x') = f(x) - f(x')$

Suppose the model is linear. Then:

$$\sum_i \phi_i(f, x, x') = \sum_i f(x) - f(x + (x' - x)e_i) = \sum_i f(x - x')e_i = f(x) - f(x')$$

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What goes wrong when the model is not linear? Consider $f(x) = x_1x_2$ for $x' = (0,0)$ and $x = (1,1)$.

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What goes wrong when the model is not linear? Consider $f(x) = x_1 x_2$ for $x' = (0,0)$ and $x = (1,1)$.

We then have:

$$\begin{aligned} f(x) - f(x') &= 1 \\ \phi_1(f, x, x') &= f((1,1)) - f((0,1)) = 1 \\ \phi_2(f, x, x') &= f((1,1)) - f((1,0)) = 1 \end{aligned}$$

$$\sum_i \phi_i(f, x, x') = 2 \neq f(x) - f(x')$$

Integrated Gradients

Define feature attributions $\phi_i(f, x, x')$ for model f , input x , and a reference input x' as follows:

$$\phi_i(f, x, x') = (x_i - x'_i) \int_0^1 \frac{\partial f(x' + \alpha(x - x'))}{\partial x_i} d\alpha$$

Consider our previous example: $f(\mathbf{x}) = x_1 x_2$ for $\mathbf{x}' = (0,0)$ and $\mathbf{x} = (1,1)$
(on the board)

Interpretation: We are integrating the i th component of the gradient along the straight-line path from $\mathbf{x}'$ to $\mathbf{x}$, then scaling by the displacement $x_i - x'_i$.

Contrast this with evaluating the gradient at x only (saliency maps).

Integrated Gradients

We must select a baseline. Common choices are:

- A black image $x' = \mathbf{0}$
- The average image $x' = E[X]$
- A blurred version of x

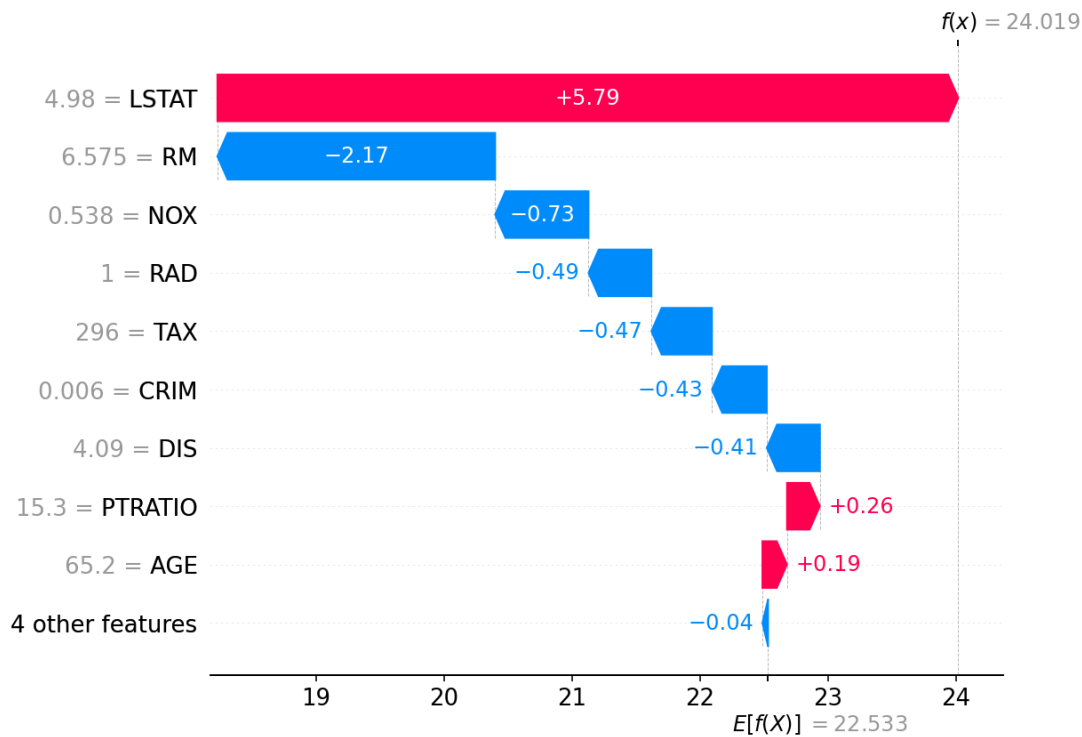
Expected Gradients

- Sometimes there is a natural way to define a baseline x'
- But this is not always the case
- Expected gradients avoids the need to define x'
- Intuition: take expected integrated gradients when drawing the baseline from the data distribution

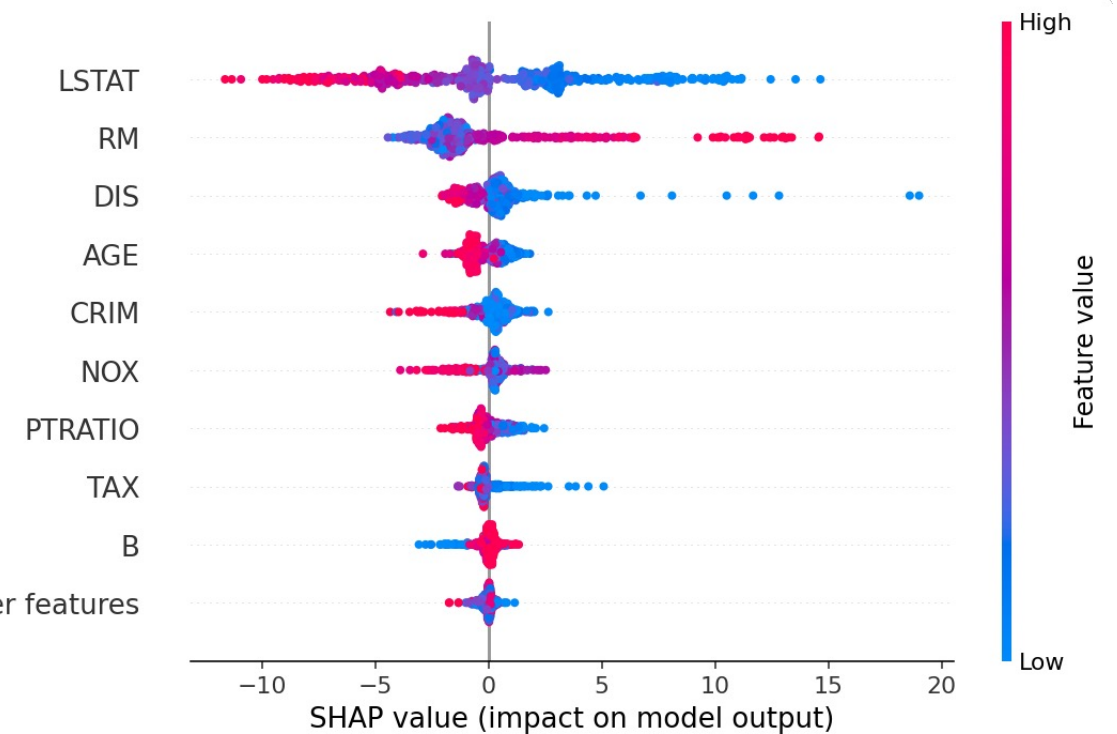
$$\phi_i^{EG}(x) = \mathbb{E}_{X'}[\phi_i^{IG}(x, X')]$$

SHAP Values: a game-theoretic approach

For one prediction



For all predictions



Summary

- Interpretable and explainable are related but distinct ideas
- The importance of interpretability is argued and views are changing
- Neural networks are not *interpretable*, but we have techniques that can quantify the effect of features on each individual prediction – in other words, *explain* the prediction
- Saliency maps and Grad-CAM are convenient but have limitations
- Integrated gradients, expected gradients, and SHAP are more principled but also more computationally intensive